

AI2 Exercise 2a

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Representing the Circuit in Boolean Logic

Representing the Circuit in Boolean Logic

SD =

$$\begin{aligned} &(\neg Ab_{G1} \rightarrow ((\neg A \vee B) \leftrightarrow G_1)) \wedge \\ &(\neg Ab_{G2} \rightarrow ((A \vee \neg B) \leftrightarrow G_2)) \wedge \\ &(\neg Ab_{G3} \rightarrow ((\neg C \vee D) \leftrightarrow G_3)) \wedge \\ &(\neg Ab_{G4} \rightarrow ((C \vee \neg D) \leftrightarrow G_4)) \wedge \\ &(\neg Ab_{G5} \rightarrow ((G_1 \wedge G_2) \leftrightarrow G_5)) \wedge \\ &(\neg Ab_{G6} \rightarrow ((G_3 \wedge G_4) \leftrightarrow G_6)) \wedge \\ &(\neg Ab_{G7} \rightarrow ((\neg G_5 \vee \neg G_6) \leftrightarrow G_7)) \end{aligned}$$

CNF =

$$\begin{aligned} &(Ab_{G1} \vee A \vee G_1) \wedge (Ab_{G1} \vee \neg B \vee G_1) \wedge (Ab_{G1} \vee \neg G_1 \vee \neg A \vee B) \wedge \\ &(Ab_{G2} \vee \neg A \vee G_2) \wedge (Ab_{G2} \vee B \vee G_2) \wedge (Ab_{G2} \vee \neg G_2 \vee A \vee \neg B) \wedge \\ &(Ab_{G3} \vee C \vee G_3) \wedge (Ab_{G3} \vee \neg D \vee G_3) \wedge (Ab_{G3} \vee \neg G_3 \vee \neg C \vee D) \wedge \\ &(Ab_{G4} \vee \neg C \vee G_4) \wedge (Ab_{G4} \vee D \vee G_4) \wedge (Ab_{G4} \vee \neg G_4 \vee C \vee \neg D) \wedge \\ &(Ab_{G5} \vee \neg G_1 \vee \neg G_2 \vee G_5) \wedge (Ab_{G5} \vee \neg G_5 \vee G_1) \wedge (Ab_{G5} \vee \neg G_5 \vee G_2) \wedge \\ &(Ab_{G6} \vee \neg G_3 \vee \neg G_4 \vee G_6) \wedge (Ab_{G6} \vee \neg G_6 \vee G_3) \wedge (Ab_{G6} \vee \neg G_6 \vee G_4) \wedge \\ &(Ab_{G7} \vee G_5 \vee G_7) \wedge (Ab_{G7} \vee G_6 \vee G_7) \wedge (Ab_{G7} \vee \neg G_7 \vee \neg G_5 \vee \neg G_6) \end{aligned}$$

Minimum-Cardinality Diagnoses

“How many models are there for SD ? (without any observations)?”:

There are 34992 satisfying assignments of SD without any variable evidence fixed.

“How many models are there for $SD \wedge Obs_1$?”: There are 12 satisfying assignments of $SD \wedge Obs_1$. In this solution the unit clauses of Obs_1 were added.

“Write down all minimum-cardinality diagnoses for $(SD, Comps)$ with Obs_2 (using your MaxSAT formalization).”: Minimum cardinality = 1
Minimum cardinality diagnoses:

$$\{\{G_1\}, \{G_2\}, \{G_3\}, \{G_4\}, \{G_5\}, \{G_6\}, \{G_7\}\}$$

“Give one minimum-cardinality diagnosis for $(SD, Comps)$ with Obs_3 .”:
Minimum cardinality = 2 Minimum cardinality diagnosis:

$$\{G_3, G_5\}$$